

Equation of motion

$$h[k+1] = h[k] + a_h[k]\{f_{dh}[k] + f_T[k]\} + \delta h_D[k],$$

Motion error

$$\delta h[k+1] = \delta h[k] + \Delta h_d[k] - a_h[k]\{f_{dh}[k] + f_T[k]\} - \delta h_D[k],$$

$\delta h[k] = h_d[k] - h[k]$, $h_d[k]$ is the desired motion trajectory, $\delta h_D[k]$ is the motion error caused by the disturbance, and $\Delta h_d[k] = h_d[k+1] - h_d[k]$ is the desired motion step size. Radius R enters through $\bar{h} = h/R$; hence $\Delta h_d = R \Delta \bar{h}_d$ and $d/dh = (1/R) d/d\bar{h}$.

Measurement

(d -step delay)

$$\delta h_m[k] = \delta h[k-d] + n_h[k],$$

Control objective: $\delta h[k+1] = \lambda_c \cdot \delta h[k]; \quad 0 \leq \lambda_c < 1$

Ideal control effort (no measurement delay):

$$\bar{f}_{dh}[k] = a_h^{-1}[k]\{\Delta h_d[k] + (1 - \lambda_c) \delta h[k] - \delta h_D[k]\}$$

Motion error:

$$\delta h[k+1] = \delta h[k] + \Delta h_d[k] - a_h[k]\{\bar{f}_{dh}[k] + f_T[k]\} - \delta h_D[k] = \lambda_c \cdot \delta h[k] - a_h[k]f_T[k]$$

d -step measurement delay:

$$\delta h[k] = \delta h[k-d] + \sum_{i=1}^d \Delta h_d[k-i] - \sum_{i=1}^d a_h[k-i]\{f_{dh}[k-i] + f_T[k-i]\} - \sum_{i=1}^d \delta h_D[k-i]$$

Ideal control effort (with d -step measurement delay):

$$\begin{aligned} \bar{f}_{dh}[k] = a_h^{-1}[k] \Big\{ & \underbrace{\left[\Delta h_d[k] + (1 - \lambda_c) \sum_{i=1}^d \Delta h_d[k-i] \right]}_{\Delta h_d^d[k]} + (1 - \lambda_c) \left[\delta h[k-d] - \sum_{i=1}^d a_h[k-i] f_{dh}[k-i] \right] \\ & - \underbrace{\left[\delta h_D[k] + (1 - \lambda_c) \sum_{i=1}^d \delta h_D[k-i] \right]}_{\delta h_D^d[k]} \Big\} \end{aligned}$$

Motion error:

$$\delta h[k+1] = \lambda_c \cdot \delta h[k] - \underbrace{\left\{ a_h[k]f_T[k] + (1 - \lambda_c) \sum_{i=1}^d a_h[k-i]f_T[k-i] \right\}}_{\delta h_T^d[k]}$$

Implementable control effort

$(\delta \hat{h}_D^d \equiv 0$: disturbance not estimated)

$$f_{dh}[k] = \hat{a}_h^{-1}[k] \left\{ \Delta h_d^d[k] + (1 - \lambda_c) \left[\delta h_m[k] - \sum_{i=1}^d \hat{a}_h[k-i] f_{dh}[k-i] \right] \right\}$$

Motion error

$$\delta h[k+1] = \lambda_c \cdot \delta h[k] - a_h[k] [f_{dh}[k] - \bar{\bar{f}}_{dh}[k]] - \delta h_T^d[k]$$

$$a_h[k] \bar{\bar{f}}_{dh}[k] = \Delta h_d^d[k] + (1 - \lambda_c) \left[\delta h[k-d] - \sum_{i=1}^d a_h[k-i] f_{dh}[k-i] \right] - \delta h_D^d[k]$$

$$a_h[k] f_{dh}[k] = (1 + e_{a_h}[k] \hat{a}_h^{-1}[k]) \left\{ \Delta h_d^d[k] + (1 - \lambda_c) \left[(\delta h[k-d] + n_h[k]) - \sum_{i=1}^d \hat{a}_h[k-i] f_{dh}[k-i] \right] \right\}$$

$$\begin{aligned} a_h[k] f_{dh}[k] - a_h[k] \bar{\bar{f}}_{dh}[k] &= e_{a_h}[k] f_{dh}[k] + (1 - \lambda_c) n_h[k] \\ &\quad + (1 - \lambda_c) \sum_{i=1}^d (a_h[k-i] - \hat{a}_h[k-i]) f_{dh}[k-i] + \delta h_D^d[k] \end{aligned}$$

Gain model (definitions only; $a_o = a_{nom}$ = far-field gain, $\bar{h}_d = h_d/R$):

$$\begin{aligned} a_h[k] &= a_h(h_d[k]) - a'_h[k] \delta h[k], & a_h(h_d[k+1]) &= a_h(h_d[k]) + a'_h[k] \Delta h_d[k], \\ a'_h[k] &:= \left. \frac{da_h}{dh} \right|_{h_d} = \frac{s_h[k]}{R}, & s_h[k] &:= \left. \frac{da_h}{d\bar{h}} \right|_{h_d} = \frac{p[k]}{\bar{h}_d - 1} \frac{a_h[k](a_o - a_h[k])}{a_o}. \end{aligned}$$

Radius R

(one bridge) $\bar{h}_d = h_d/R$ is the only place R enters. The slope has a physical and an $\bar{h}$ face, $a'_h = s_h/R$ with $s_h := da_h/d\bar{h}$ (R -free). In a product $a'_h \cdot L$: for the trajectory step $L = \Delta h_d = R \Delta \bar{h}_d$ the R cancels ($a'_h \Delta h_d = s_h \Delta \bar{h}_d$); for a tracking-error length $L \in \{\delta h, F_{dh} e_{a_h}, \varepsilon_h\}$ the $1/R$ stays ($a'_h L = (s_h/R)L$).

Gain-error back-step ($O(\Delta h_d)$ slope-error correction dropped):

$$a_h[k-i] - \hat{a}_h[k-i] \approx e_{a_h}[k] - (h_d[k] - h_d[k-i]) (a'_h[k] - \hat{a}'_h[k]) \approx e_{a_h}[k]$$

$$a_h[k] f_{dh}[k] - a_h[k] \bar{\bar{f}}_{dh}[k] \approx e_{a_h}[k] \underbrace{\left\{ f_{dh}[k] + (1 - \lambda_c) \sum_{i=1}^d f_{dh}[k-i] \right\}}_{F_{dh}[k]} + \delta h_D^d[k] + (1 - \lambda_c) n_h[k]$$

$$\delta h[k+1] = \lambda_c \cdot \delta h[k] - F_{dh}[k] \cdot e_{a_h}[k] - \delta h_T^d[k] - \underbrace{\left\{ \delta h_T^d[k] + (1 - \lambda_c) n_h[k] \right\}}_{\varepsilon_h[k]}$$

True gain recursion

(exact curve swept to $h_d[k+1]$; leading order; a'_h expanded)

$$\boxed{\begin{aligned} a_h[k+1] &= a_h + s_h[k] \Delta \bar{h}_d + \frac{s_h[k]}{R} [(1-\lambda_c) \delta h + F_{dh} e_{a_h} + \varepsilon_h], & s_h[k] &= \frac{p}{\bar{h}_d - 1} \frac{a_h(a_o - a_h)}{a_o} \\ p[k+1] &= p \end{aligned}}$$

True state equation

($\delta h_D^d \equiv 0$; leading order)

$$\begin{aligned} \delta h_1[k+1] &= \delta h_2[k] \\ \delta h_2[k+1] &= \delta h_3[k] \\ \delta h_3[k+1] &= \lambda_c \delta h_3[k] - F_{dh}[k] e_{a_h}[k] - \varepsilon_h[k] \\ a_h[k+1] &= a_h[k] + s_h[k] \Delta \bar{h}_d + \frac{s_h[k]}{R} [(1-\lambda_c) \delta h_3 + F_{dh} e_{a_h} + \varepsilon_h] \\ p[k+1] &= p[k] \end{aligned}$$

State vector: $\mathbf{x}[k] = [\delta h_1, \delta h_2, \delta h_3, a_h, p]^\top[k]$.

Output equation

(a_{hm} observes $a_h[k-d]$; leading-order delay back-off, $\Delta \bar{H}_d = (h_d[k] - h_d[k-d])/R$)

$$\begin{cases} y_1[k] = \delta h_m[k] = \delta h_1[k] + \underbrace{n_h[k]}_{r_1[k]} \\ y_2[k] = \underbrace{a_h[k-d] + n_a[k]}_{a_{hm}[k]} \approx a_h[k] - s_h[k] \Delta \bar{H}_d[k] + \underbrace{n_a[k]}_{r_2[k]} \end{cases}$$

$$\mathbf{y}[k] = \mathbf{H} \mathbf{x}[k] + \mathbf{r}[k], \quad \mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & -\Delta \bar{H}_d \frac{s_h[k]}{p} \end{bmatrix}$$

y_2 observes a_h directly ($H_{24}=1$; $O(\Delta \bar{H}_d)$ correction dropped); p enters only through the delay, $H_{25} = -\Delta \bar{H}_d s_h[k]/p$.
Curvature bias $\frac{1}{2} a_h'' \sigma_h^2$ in a_{hm} folded into r_2/R_2 (deferred).

Estimator (model)

($\hat{a}_h$ swept by $\hat{s}_h[k]$; $\hat{p}$ random walk)

$$e_{\delta h_1} = \delta h_1 - \hat{\delta h}_1, \quad e_{a_h} = a_h - \hat{a}_h, \quad e_p = p - \hat{p}$$

$$e_{y_1} = e_{\delta h_1} + \underbrace{n_h}_{r_1}, \quad e_{y_2} = a_{hm} - (\hat{a}_h - \Delta \bar{H}_d \hat{s}_h[k]) = e_{a_h} - \Delta \bar{H}_d \frac{\hat{s}_h[k]}{\hat{p}} e_p + \underbrace{n_a}_{r_2}$$

$$\delta \hat{h}_1[k+1] = \delta \hat{h}_2 + \ell_{11} e_{y_1} + \ell_{12} e_{y_2}$$

$$\delta \hat{h}_2[k+1] = \delta \hat{h}_3 + \ell_{21} e_{y_1} + \ell_{22} e_{y_2}$$

$$\delta \hat{h}_3[k+1] = \lambda_c \delta \hat{h}_3 + \ell_{31} e_{y_1} + \ell_{32} e_{y_2}$$

$$\hat{a}_h[k+1] = \hat{a}_h + \hat{s}_h[k] \Delta \bar{h}_d + (1-\lambda_c) \frac{\hat{s}_h[k]}{R} \delta \hat{h}_3 + \ell_{41} e_{y_1} + \ell_{42} e_{y_2}$$

$$\hat{p}[k+1] = \hat{p} + \ell_{51} e_{y_1} + \ell_{52} e_{y_2}$$

Error dynamics

(True – Estimator; leading order, $\mathbf{Lr}$ dropped. $s_a[k] := \partial s_h / \partial a_h = \frac{p}{h_d - 1} \frac{a_o - 2a_h}{a_o}$, $\partial s_h / \partial p = s_h[k]/p$)

$$\begin{aligned}
e_{\delta h_1}[k+1] &= e_{\delta h_2} - \ell_{11} e_{\delta h_1} - \ell_{12} e_{a_h} + \ell_{12} \Delta \bar{H}_d \frac{s_h[k]}{p} e_p \\
e_{\delta h_2}[k+1] &= e_{\delta h_3} - \ell_{21} e_{\delta h_1} - \ell_{22} e_{a_h} + \ell_{22} \Delta \bar{H}_d \frac{s_h[k]}{p} e_p \\
e_{\delta h_3}[k+1] &= \lambda_c e_{\delta h_3} - F_{dh} e_{a_h} - \underbrace{\varepsilon_h}_{q_3} - \ell_{31} e_{\delta h_1} - \ell_{32} e_{a_h} + \ell_{32} \Delta \bar{H}_d \frac{s_h[k]}{p} e_p \\
e_{a_h}[k+1] &= \left(1 + \Delta \bar{h}_d s_a[k] + \frac{s_h[k]}{R} F_{dh}\right) e_{a_h} + (1 - \lambda_c) \frac{s_h[k]}{R} e_{\delta h_3} \\
&\quad + \Delta \bar{h}_d \frac{s_h[k]}{p} e_p + \underbrace{\frac{s_h[k]}{R} \varepsilon_h}_{q_4} - \ell_{41} e_{\delta h_1} - \ell_{42} e_{a_h} + \ell_{42} \Delta \bar{H}_d \frac{s_h[k]}{p} e_p \\
e_p[k+1] &= e_p - \ell_{51} e_{\delta h_1} - \ell_{52} e_{a_h} + \ell_{52} \Delta \bar{H}_d \frac{s_h[k]}{p} e_p
\end{aligned}$$

$$\mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dh} & 0 \\ 0 & 0 & (1 - \lambda_c) \frac{s_h[k]}{R} & 1 + \Delta \bar{h}_d s_a[k] + \frac{s_h[k]}{R} F_{dh} & \Delta \bar{h}_d \frac{s_h[k]}{p} \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{e}[k+1] = \mathbf{F}_e[k] \mathbf{e}[k] - \mathbf{L}[k] \mathbf{H} \mathbf{e}[k] + \mathbf{q}[k]$$

$\mathbf{q} = [0, 0, q_3, q_4, 0]^\top$: $q_3 = -\varepsilon_h$, $q_4 = (s_h[k]/R)\varepsilon_h = a'_h[k]\varepsilon_h$ (rank-1 gain block, fully correlated $q_4 = -a'_h[k]q_3$); Q values below. Leading order keeps each channel's leading term ($F_{e,44}$'s $\Delta \bar{h}_d s_a$, and p 's sole plant path $F_{e,45} = \Delta \bar{h}_d s_h[k]/p$) but drops the same-order correction to the already- $O(1)$ $H_{24} = 1$.

Process noise Q

(Path C strict: only the current white thermal drives Q ; the ε_h MA(d) tail $\rightarrow$ closed-loop variance, $n_h \rightarrow R_1$) $\varepsilon_h = \delta h_T^d + (1 - \lambda_c)n_h$ mixes the thermal MA(d) tail and sensor noise; a valid KF Q keeps only the *current* white thermal step $a_h f_T[k]$, with $\text{Var}(a_h f_T) = a_h^2 \sigma_{f_T}^2 = 4k_B T a_h[k]$ (fluctuation–dissipation: $\sigma_{f_T}^2 = 4k_B T \gamma / \Delta t$, $a_h = \Delta t / \gamma$). With $q_3 = -a_h f_T[k]$, $q_4 = a'_h a_h f_T[k]$:

$$\begin{aligned}
Q_{33}[k] &= 4k_B T a_h[k], & Q_{44}[k] &= a'_h[k]^2 Q_{33}[k], & Q_{34}[k] &= -a'_h[k] Q_{33}[k], \\
Q_{55} &= 0 \quad (p \text{ constant; optional floor } \sigma_p^2 \text{ if slow drift allowed}).
\end{aligned}$$

The MA-tail inflation of σ_{th}^2 (past thermal, cross-step correlated) is the standard correlated-noise KF trade-off, carried by the closed-loop variance, not Q (as in the baseline Path C).

Measurement noise R

($\mathbf{R} = \text{diag}(R_1, R_2)$) $R_1 = \sigma_{n_h}^2$ (sensor noise on δh_m). a_{hm} is the IIR variance-ratio gain readout (paper 2025 Eq.9–13): from the random tracking-error component δh_r ,

$$\hat{\sigma}_{\delta h_r}^2[k] = C_{dpmr} 4k_B T a_h[k] + C_n \sigma_{n_h}^2, \quad C_{dpmr} = 2 + \frac{1}{1 - \lambda_c^2}, \quad C_n = \frac{2}{1 + \lambda_c},$$

$$a_{hm}[k] = \frac{\hat{\sigma}_{\delta h_r}^2[k] - C_n \sigma_{n_h}^2}{C_{dpmr} 4k_B T}.$$

The EWMA (coefficient a_{cov}) forming $\hat{\sigma}_{\delta h_r}^2$ propagates a χ^2 variance $\text{Var}(\hat{\sigma}_{\delta h_r}^2) = a_{cov} \text{IF}_{var}(\sigma_{\delta h_r}^2)^2$ ($\text{IF}_{var} = \text{MA}(2)$ inflation), so the linear inversion gives

$$R_2[k] = a_{cov} \text{IF}_{var} (a_h[k] + \xi)^2, \quad \xi := \frac{C_n}{C_{dpmr}} \frac{\sigma_{n_h}^2}{4k_B T}.$$

Ported from the baseline R_2 chain (identical IIR readout, $a_x \rightarrow a_h$). Deferred additions, both small, folded into R_2 later: (i) the a_{hm} curvature/Jensen bias $\frac{1}{2}a_h''\sigma_h^2$ from the thermal excursion σ_h^2 ; (ii) the d -step accumulation of q_4 ($\sim \sum Q_{44}$, the delay-propagation analogue of the baseline's $+5Q_{77}$).



